



RICARDO LOPEZ

Aspiring Data Science Engineering

About Me

I'm Ricardo Lopez, a passionate B.Tech Electronics and Communication Engineering student at SRM Institute of Science and Technology (SRMIST), driven by curiosity and a desire to build impactful technology.



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[Ricardo Lopez](#)
([Linkedin Profile](#)).

Language

- English
- Tamil
- Telugu (Basic)

Expertise

- Data Cleaning & Data Preparation
- Exploratory Data Analysis (EDA)
- Data Visualization
- SQL & Spreadsheet Proficiency
- Analytical Thinking & Communication
- Leadership Skills
- Punctuality
- Communication
- Problem Solving

Education

B.Tech in Electronics and Communication with Data Science SRMIST , Ramapuram (2024 - Present)

Pursuing a B.Tech in Electronics and Communication Engineering with a strong foundation in circuit design, signal processing, and telecommunication principles. Actively engaging in coursework and lab projects focused on emerging technologies, including IoT and embedded systems.

Higher Secondary Education

Montfort . Mat . Hr . Sec . School (2022 - 2024)

Completed Higher Secondary Education with a focus on Mathematics, Physics, and Chemistry, achieving a strong academic foundation.

Achievements

Secured 1st Place (out of 75+ teams) at the National Level Hackathon organized by St. Joseph College of Engineering.

Secured Track Prize at Protothon , National Level Hackathon organized by HITS , Chennai

Secured 2nd Place (out of 30+ teams) at the Paper Presentation Competition organized by SIMATS

Projects

Field - Eye (Aug 2025 - Present) **Description:**

Field-Eye-1.0 is an autonomous agricultural rover built on an ESP32 ecosystem, designed for real-time crop surveillance and early detection of pests using live video streams and onboard AI. Equipped with ESP32-CAM and GPS, the rover autonomously navigates farmland in a grid pattern, capturing both visual and location data. This data is streamed live and processed by an embedded or connected AI model to detect crop stress or pest presence. When abnormalities are flagged, the system alerts the user with precise grid coordinates for timely intervention

VitalDrive: A Heart Rate-Monitored Vehicle Safety System (Jul 2025 - Present) **Description :**

An IoT safety system that proactively prevents accidents caused by driver medical emergencies like fainting or cardiac arrest. It uses a non-invasive sensor to continuously monitor the driver's heart rate in real-time. If a critical drop is detected, VitalDrive automatically initiates a fail-safe protocol: gradually stopping the vehicle, activating hazard lights, and sending an SOS alert with GPS coordinates to emergency services.

SpinGen 2.0 (Jul 2025 - Present) **Description :**

Evolving my award-winning research paper, this project develops a functional hardware prototype to harvest wasted kinetic energy from the ambient motion of fans. It uses an optimized magnet-coil induction system to generate electricity from a non-invasive, adaptable device.